

Feature Engineering

```
# =====  
# 03_feature_engineering.R  
# Airline Service Performance Visual Analytics  
# =====  
  
# 1. LOAD PACKAGE -----  
  
library(dplyr)  
  
# 2. FILE PATHS -----  
  
input_path <- "D:/data-portfolio/08_airline_service_performance_visual_analytics/data/processed/airline.  
output_path <- "D:/data-portfolio/08_airline_service_performance_visual_analytics/data/processed/airlin  
  
# 3. READ CLEAN DATA -----  
  
clean_data <- read.csv(  
  input_path,  
  stringsAsFactors = FALSE,  
  na.strings = c("", "NA")  
)  
  
cat("\n=== CLEAN DATA IMPORTED ===\n")
```

```
##  
## === CLEAN DATA IMPORTED ===
```

```
cat(  
  "Number of rows:",  
  nrow(clean_data),  
  "\n"  
)
```

```
## Number of rows: 80972
```

```
cat(  
  "Number of columns:",  
  ncol(clean_data),  
  "\n"  
)
```

```
## Number of columns: 17
```

```
# 4. RESTORE DATE FORMAT -----
```

```
clean_data$Month <- as.Date(  
  clean_data$Month  
)
```

```
cat("\n=== DATE CHECK ===\n")
```

```
##
```

```
## === DATE CHECK ===
```

```
cat(  
  "Date range:",  
  as.character(min(clean_data$Month)),  
  "to",  
  as.character(max(clean_data$Month)),  
  "\n"  
)
```

```
## Date range: 2010-01-01 to 2024-02-01
```

```
# 5. CREATE TIME VARIABLES -----
```

```
analysis_data <- clean_data %>%  
  mutate(  
    Year = as.integer(  
      format(Month, "%Y")  
    ),  
  
    Month_Number = as.integer(  
      format(Month, "%m")  
    ),  
  
    Month_Name = month.abb[  
      Month_Number  
    ]  
  )
```

```
cat("\n=== TIME VARIABLES ===\n")
```

```
##
```

```
## === TIME VARIABLES ===
```

```
cat(  
  "Number of years:",  
  n_distinct(analysis_data$Year),  
  "\n"  
)
```

```
## Number of years: 15
```

```
cat(  
  "First year:",  
  min(analysis_data$Year),  
  "\n"  
)
```

```
## First year: 2010
```

```
cat(  
  "Last year:",  
  max(analysis_data$Year),  
  "\n"  
)
```

```
## Last year: 2024
```

```
cat(  
  "Month numbers:",  
  paste(  
    sort(  
      unique(analysis_data$Month_Number)  
    ),  
    collapse = ", "  
  ),  
  "\n"  
)
```

```
## Month numbers: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
```

```
cat(  
  "Month names:",  
  paste(  
    month.abb,  
    collapse = ", "  
  ),  
  "\n"  
)
```

```
## Month names: Jan, Feb, Mar, Apr, May, Jun, Jul, Aug, Sep, Oct, Nov, Dec
```

```
# 6. CREATE ANALYSIS LEVEL -----
```

```
analysis_data <- analysis_data %>%  
  mutate(  
    Analysis_Level = case_when(  
      Route == "All Ports-All Ports" &  
        Airline == "All Airlines" ~  
        "Network",
```

```

    Route == "All Ports-All Ports" &
      Airline != "All Airlines" ~
        "Airline",

    Route != "All Ports-All Ports" &
      Airline == "All Airlines" ~
        "Route",

    TRUE ~
      "Airline Route"
  )
)

```

```
cat("\n=== ANALYSIS LEVELS ===\n")
```

```
##
## === ANALYSIS LEVELS ===
```

```

analysis_level_summary <- analysis_data %>%
  count(
    Analysis_Level
  )

print(
  analysis_level_summary
)

```

```

##   Analysis_Level      n
## 1      Airline  1174
## 2   Airline Route 59472
## 3      Network   170
## 4        Route 20156

```

```

cat(
  "Total rows across analysis levels:",
  sum(analysis_level_summary$n),
  "\n"
)

```

```
## Total rows across analysis levels: 80972
```

```
# 7. CREATE THE FOUR ANALYTICAL LAYERS -----
```

```

network_month <- analysis_data %>%
  filter(
    Analysis_Level == "Network"
  )

```

```
airline_month <- analysis_data %>%
```

```

filter(
  Analysis_Level == "Airline"
)

route_month <- analysis_data %>%
  filter(
    Analysis_Level == "Route"
  )

airline_route_month <- analysis_data %>%
  filter(
    Analysis_Level == "Airline Route"
  )

cat("\n=== ANALYTICAL LAYER ROW COUNTS ===\n")

```

```

##
## === ANALYTICAL LAYER ROW COUNTS ===

```

```

cat(
  "Network-month rows:",
  nrow(network_month),
  "\n"
)

```

```

## Network-month rows: 170

```

```

cat(
  "Airline-month rows:",
  nrow(airline_month),
  "\n"
)

```

```

## Airline-month rows: 1174

```

```

cat(
  "Route-month rows:",
  nrow(route_month),
  "\n"
)

```

```

## Route-month rows: 20156

```

```

cat(
  "Airline-route-month rows:",
  nrow(airline_route_month),
  "\n"
)

```

```
## Airline-route-month rows: 59472
```

```
# 8. CHECK NETWORK-MONTH STRUCTURE -----
```

```
network_month_check <- network_month %>%  
  count(  
    Month  
  ) %>%  
  filter(  
    n != 1  
  )
```

```
cat("\n=== NETWORK-MONTH CHECK ===\n")
```

```
##
```

```
## === NETWORK-MONTH CHECK ===
```

```
cat(  
  "Network months:",  
  n_distinct(network_month$Month),  
  "\n"  
)
```

```
## Network months: 170
```

```
cat(  
  "Months without exactly one network record:",  
  nrow(network_month_check),  
  "\n"  
)
```

```
## Months without exactly one network record: 0
```

```
# 9. RECALCULATE ROW-LEVEL PERFORMANCE -----
```

```
metric_check <- analysis_data %>%  
  mutate(  
  
    Calculated_Departure_OTP = ifelse(  
      Sectors_Flown > 0,  
      Departures_On_Time /  
      Sectors_Flown * 100,  
      NA  
    ),  
  
    Calculated_Arrival_OTP = ifelse(  
      Sectors_Flown > 0,  
      Arrivals_On_Time /  
      Sectors_Flown * 100,  
      NA  
    ),
```

```

    Calculated_Cancellation_Rate = ifelse(
      Sectors_Scheduled > 0,
      Cancellations /
        Sectors_Scheduled * 100,
      NA
    )
  )
)

# 10. COMPARE CALCULATED AND PUBLISHED PERCENTAGES -----

metric_check <- metric_check %>%
  mutate(

    Departure_OTP_Difference =
      Calculated_Departure_OTP -
      OnTime_Departures_Percent,

    Arrival_OTP_Difference =
      Calculated_Arrival_OTP -
      OnTime_Arrivals_Percent,

    Cancellation_Rate_Difference =
      Calculated_Cancellation_Rate -
      Cancellations_Percent
  )

cat("\n=== PERFORMANCE METRIC VALIDATION ===\n")

```

```

##
## === PERFORMANCE METRIC VALIDATION ===

```

```

cat(
  "Maximum Departure OTP difference:",
  max(
    abs(metric_check$Departure_OTP_Difference),
    na.rm = TRUE
  ),
  "\n"
)

```

```

## Maximum Departure OTP difference: 2.700062e-13

```

```

cat(
  "Maximum Arrival OTP difference:",
  max(
    abs(metric_check$Arrival_OTP_Difference),
    na.rm = TRUE
  ),
  "\n"
)

```

```
## Maximum Arrival OTP difference: 1.421085e-13
```

```
cat(  
  "Maximum Cancellation Rate difference:",  
  max(  
    abs(metric_check$Cancellation_Rate_Difference),  
    na.rm = TRUE  
  ),  
  "\n"  
)
```

```
## Maximum Cancellation Rate difference: 4.973799e-14
```

```
# 11. DEMONSTRATE WEIGHTED AGGREGATION -----
```

```
network_metric_example <- network_month %>%  
  summarise(  
    Total_Sectors_Flown =  
      sum(Sectors_Flown),  
    Total_Sectors_Scheduled =  
      sum(Sectors_Scheduled),  
    Total_Departures_On_Time =  
      sum(Departures_On_Time),  
    Total_Arrivals_On_Time =  
      sum(Arrivals_On_Time),  
    Total_Cancellations =  
      sum(Cancellations),  
    Weighted_Departure_OTP =  
      sum(Departures_On_Time) /  
      sum(Sectors_Flown) * 100,  
    Weighted_Arrival_OTP =  
      sum(Arrivals_On_Time) /  
      sum(Sectors_Flown) * 100,  
    Weighted_Cancellation_Rate =  
      sum(Cancellations) /  
      sum(Sectors_Scheduled) * 100,  
    Simple_Mean_Departure_OTP =  
      mean(  
        OnTime_Departures_Percent,  
        na.rm = TRUE  
      ),  
    Simple_Mean_Arrival_OTP =  
      mean(  
        OnTime_Arrivals_Percent,  
        na.rm = TRUE  
      )  
  )
```



```

      OnTime_Arrivals_Percent,
      na.rm = TRUE
    ),

    Simple_Mean_Cancellation_Rate =
      mean(
        Cancellations_Percent,
        na.rm = TRUE
      )
  )

cat("\n=== WEIGHTED AGGREGATION EXAMPLE ===\n")

```

```

##
## === WEIGHTED AGGREGATION EXAMPLE ===

```

```

print(
  network_metric_example
)

```

```

##   Total_Sectors_Flown Total_Sectors_Scheduled Total_Departures_On_Time
## 1              7078663              7268896              5733390
##   Total_Arrivals_On_Time Total_Cancellations Weighted_Departure_OTP
## 1              5655777              190233              80.99539
##   Weighted_Arrival_OTP Weighted_Cancellation_Rate Simple_Mean_Departure_OTP
## 1              79.89895              2.617082              81.42816
##   Simple_Mean_Arrival_OTP Simple_Mean_Cancellation_Rate
## 1              80.48373              3.028725

```

```

# 12. CHECK MISSING ANALYSIS LEVELS -----

```

```

missing_analysis_level <- analysis_data %>%
  filter(
    is.na(Analysis_Level)
  )

cat("\n=== ANALYSIS LEVEL VALIDATION ===\n")

```

```

##
## === ANALYSIS LEVEL VALIDATION ===

```

```

cat(
  "Rows with missing analysis level:",
  nrow(missing_analysis_level),
  "\n"
)

```

```

## Rows with missing analysis level: 0

```

```
# 13. SELECT FINAL ANALYTICAL VARIABLES -----
```

```
analysis_data <- analysis_data %>%  
  select(  
    Route,  
    Departing_Port,  
    Arriving_Port,  
    Airline,  
    Month,  
    Year,  
    Month_Number,  
    Month_Name,  
    Analysis_Level,  
    Sectors_Scheduled,  
    Sectors_Flown,  
    Cancellations,  
    Departures_On_Time,  
    Arrivals_On_Time,  
    Departures_Delayed,  
    Arrivals_Delayed,  
    OnTime_Departures_Percent,  
    OnTime_Arrivals_Percent,  
    Cancellations_Percent  
  )
```

```
# 14. FINAL ANALYSIS DATASET CHECK -----
```

```
cat("\n=== FINAL ANALYSIS DATASET ===\n")
```

```
##  
## === FINAL ANALYSIS DATASET ===
```

```
cat(  
  "Number of rows:",  
  nrow(analysis_data),  
  "\n"  
)
```

```
## Number of rows: 80972
```

```
cat(  
  "Number of columns:",  
  ncol(analysis_data),  
  "\n"  
)
```

```
## Number of columns: 19
```

```
cat(  
  "Number of analysis levels:",
```

```
n_distinct(analysis_data$Analysis_Level),
"\n"
)
```

```
## Number of analysis levels: 4
```

```
cat(
  "Number of years:",
  n_distinct(analysis_data$Year),
  "\n"
)
```

```
## Number of years: 15
```

```
cat(
  "Number of months:",
  n_distinct(analysis_data$Month),
  "\n"
)
```

```
## Number of months: 170
```

```
cat("\nFinal analysis level counts:\n")
```

```
##
## Final analysis level counts:
```

```
print(
  analysis_data %>%
    count(
      Analysis_Level
    )
)
```

```
##   Analysis_Level      n
## 1      Airline  1174
## 2 Airline Route 59472
## 3      Network   170
## 4      Route 20156
```

```
# 15. SAVE ANALYSIS DATA -----
```

```
write.csv(
  analysis_data,
  output_path,
  row.names = FALSE,
  na = ""
)
```

```
cat("\n=== FILE EXPORTED ===\n")
```

```
##  
## === FILE EXPORTED ===
```

```
cat(  
  "Analysis data saved to:",  
  output_path,  
  "\n"  
)
```

```
## Analysis data saved to: D:/data-portfolio/08_airline_service_performance_visual_analytics/data/proc
```

```
# 16. FEATURE ENGINEERING FINISHED -----
```

```
cat("\n=== FEATURE ENGINEERING FINISHED ===\n")
```

```
##  
## === FEATURE ENGINEERING FINISHED ===
```